The Energy Balance Project

Homeostatic Field Model and Local Actor-Motion Law

Foundation Document — Version 2.6 (Automated Adversarial Testing of the Guarded EBU Ledger)

This is an adversarial-search study, NOT an economy. No prices, ownership, or transferable EBU are introduced. It is a falsification stage over the frozen `v2.4.0` physics and V2.5 guarded ledger, asking one question:

Can an automated actor or coalition find action sequences that earn positive cumulative guarded EBU while causing persistent physical harm?

Result. Within the declared search space and budget, **no profitable persistent-harm exploit against the guarded ledger was found**. The search is strong enough to judge this: on the same fixture it **rediscovers a naive-ledger exploit** (positive control). A separate, honest caveat: the myopic *greedy* guarded adversary left the field worse than doing nothing on 4 of 12 random layouts — a candidate-exploit signal (persistence unverified) that the next stage should red-team directly.

1. Objective and Non-Goals

V2.5 showed a hand-written attack list could not game the guarded ledger. A finite hand list is weak evidence. V2.6 replaces it with an automated, deterministic search. No economic features are added; the frozen files (energy_balance, ebu_v22..v25) are unchanged and re-verified. All V2.6 code lives in new files so earlier results stay reproducible.

2. Method

2.1 Forced-action harness (reuses the frozen accounting)

A tick harness applies an arbitrary list of feasibility-clipped actor actions through the exact V2.5 accounting (the same _action_effect), so guarded/naive semantics are identical to V2.5 and the physical trajectory under physics selection is byte-identical to the frozen baseline (tested). Every tick also checks the conservation identity dX = S + G - D - Lambda - loss - spill.

2.2 Deterministic beam search

A beam search plans multi-tick, multi-actor sequences. Per tick each coalition actor chooses from a menu {rest} + transfers to each admissible neighbour at several quantities (small splits and full); the joint action is the product over the coalition; the beam keeps the top states by the objective. The action space thus spans neighbouring transfers, small splits, repeated/multi-tick actions, relays, cyclic/back-and-forth moves, coordinated multi-actor moves, and harvesting near or below regenerative reserves. The search is deterministic and consumes no randomness (tested), so it cannot peek at future events.

2.3 Two separated objectives

(I) an **automated adversary** maximising its own cumulative guarded EBU; (II) a **red-team** maximising cumulative physical harm subject to net EBU >= a positive threshold. These are kept apart throughout.

2.4 Exploit definition (fixed before the final runs)

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An exploit requires ALL of:
  1. coalition cumulative net guarded EBU > 0                 (profitable)
  2. physical condition persistently worse than a PAIRED
     no-action baseline (coalition rests, natural dynamics)
  3. the deterioration holds for a declared persistence window
     (every tail tick, not a one-tick blip)
  4. harm measured on PHYSICAL variables (threshold-aware
     burden B_R), never on EBU itself.
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Reported physical harm variables: final and post-shock viability, sustained recovery, unmet demand, dead regenerative sources, Allee-threshold crossings, source stock, B_R, transport dissipation. Reported EBU variables: issued credit, issued debit, net EBU, balances, concentration, plus search runtime and explored states.

3. Tests

45 tests pass: the 33 prior tests (verified inline) plus 12 new V2.6 behavior tests — observational trajectory exactly unchanged; conservation during multi-action sequences; search reproducibility; genuinely distinct random layouts; rediscovery of a naive exploit; coalition totals = issued credit - debit; a lossless restoring multi-actor cycle earns no positive EBU; splitting cannot increase reward on the same trajectory; the exploit classifier requires BOTH positive net AND persistent harm; the search uses no future randomness; physics-only identical to the frozen baseline.

4. Results

4.1 (A) Five policies from an identical initial state

Red-team fixture: a 3x3 world, an Allee source at the centre, eight deficient rim consumers, a two-actor coalition, 10 ticks. (This harsh, under-served world holds viability near 11% for every reasoned policy; it is a search fixture, not a homeostasis demo.)

Policy	viable %	B_R	dead	src stock	coalition net EBU
1 physics / no EBU	11.1	77.8	0	8.6	0.00
2 naive greedy adversary	11.1	88.9	0	12.5	+15.13
3 guarded greedy adversary	11.1	71.1	0	9.0	+21.21
4 random adversary	22.2	206.2	1	0.0	-126.69
5 automated guarded (beam)	11.1	78.2	0	8.9	+13.82

Observation: the random adversary is the only one that clearly worsens the field (B_R 206, a dead source) and it loses EBU (-126.69); the beam explored 22,743 states.

4.2 (B) Red-team search for profitable persistent harm

Ledger	verdict	net EBU	persistent harm	mean tail-harm (B_R)	states
naive (positive control)	EXPLOIT FOUND	+15.35	yes	+102.91	3,809
guarded	no exploit found	+0.91	no	-5.47	10,971

Experimental result: the search rediscovers a naive exploit — the coalition harvests the Allee source below its threshold to serve deficient consumers, earning positive naive credit (plain-burden improvement) at zero naive cost while the source dies; harm persists through the whole tail. Under the guarded ledger the same search finds no profitable persistent harm: the best net-positive sequence it finds actually leaves the field better than no action (mean tail-harm -5.47).

Interpretation: the positive control confirms the search is strong enough to expose a known ledger weakness, so its failure to break the guarded ledger on this fixture is meaningful — but see the limitations before generalising.

4.3 (C) Randomized-layout topology study

Guarded *greedy* adversary vs a paired no-action baseline, on 12 genuinely distinct random 5x5 Allee layouts (30 ticks each). Harm = final B_R(attack) - B_R(no-action); positive means the adversary left the field worse than doing nothing.

Metric	mean	sd	min	max
coalition net guarded EBU	530.24	130.47	387.41	819.94
final viability %	58.67	32.63	0.00	100.00
harm: B_R(attack) - B_R(no-action)	-77.02	249.36	-271.13	+592.15

Experimental result: on average the greedy guarded adversary is net-helpful (mean harm -77, i.e. better than no action), BUT on **4 of 12** layouts it is net-harmful (harm up to +592) while net EBU is positive.

Interpretation / caveat: this is a candidate-exploit signal for the MYOPIC greedy policy — not a confirmed exploit, because persistence was not checked with the tail window here. The red-team in 4.2 tested only the small hand-built fixture, not these layouts.

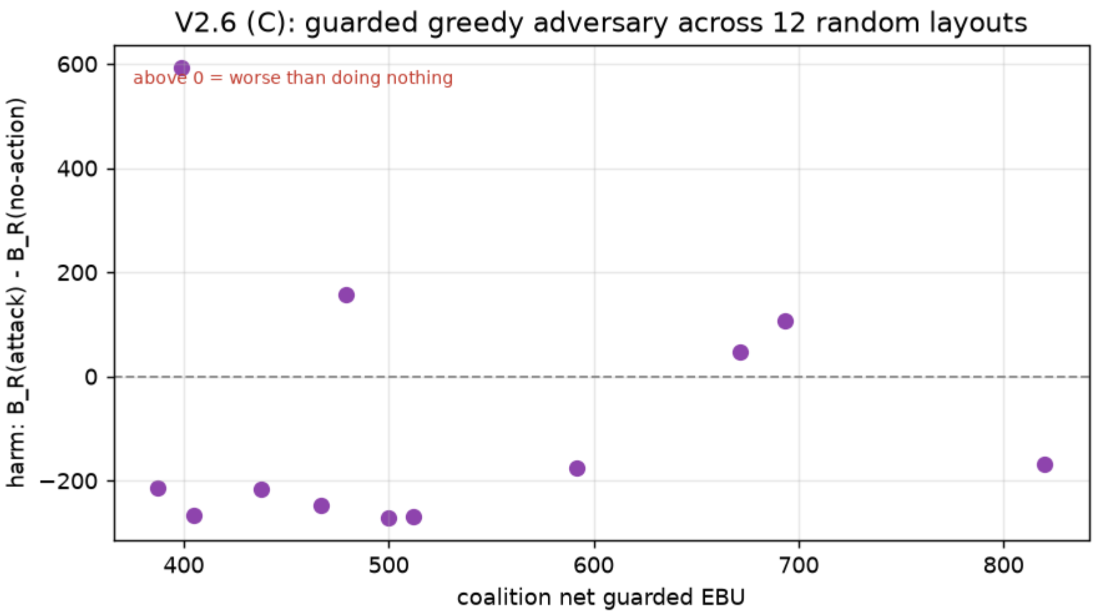

Figure 1. Guarded greedy adversary across 12 random layouts. Most points sit below zero (better than no action), but four sit above it (worse than no action) with positive net EBU.

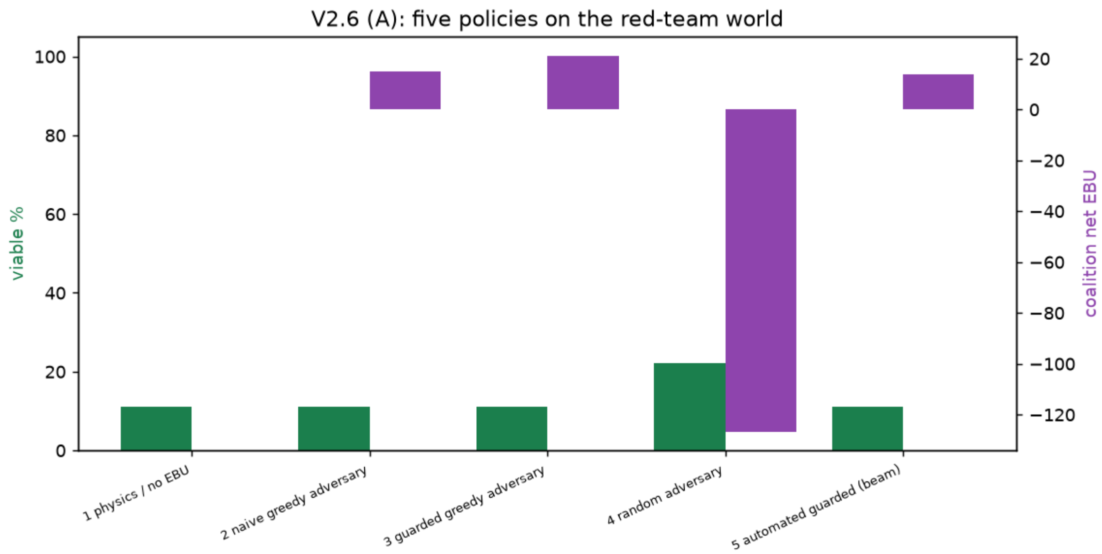

Figure 2. Five policies on the red-team fixture: viability (green) and coalition net EBU (purple). Only the random adversary worsens the field, and it loses EBU.

5. Verdict

No profitable persistent-harm exploit against the guarded ledger was found within the declared search space and computational budget. This is a falsification result, not a security proof: we do NOT claim guarded EBU is secure or that gaming is impossible. The claim is bounded by the fixture, the action menu, the beam depth/width, and the coalition size actually searched.

The strongest honest caveat is finding (C): the myopic greedy guarded adversary left the field worse than no action on 4 of 12 random layouts while earning positive EBU. The proper exploit definition (persistence window) was not applied there, so these are signals, not confirmed exploits — but they show the guarded incentive does not provably align with health on every topology, and they define the top priority for the next stage.

Hypothesis (to test next): the greedy harm arises because guarded credit is per-action and local; a myopic actor can take locally-credited actions whose combined multi-tick effect is worse than inaction. A lookahead or a persistence-aware debit may close it.

6. Limitations and Next Steps

1. The red-team searched one small hand-built fixture and a 2-actor coalition; it did not red-team the harmful random layouts from (C).
2. Beam search is incomplete; a negative result reflects the searched space, not all sequences.
3. Harm in (C) is a single-point comparison without a persistence window.
4. Coalition size, action-menu granularity, and depth/width were kept small for runtime.

Next: apply the full exploit definition (persistence) to the (C) harmful layouts; grow the coalition and action menu; add a learning/local-optimiser adversary; and if a guarded exploit is confirmed, implement a corrected guarded variant and compare old vs corrected on the same attacks and physical baselines, checking the fix does not reward hoarding or starve demand.

Reproducibility. Baseline commit, branch commit, exact commands, Python version, dependencies, seeds, and all search/experiment parameters are recorded in results/v2.6/MANIFEST.md; the captured final output is results/v2.6/v26_experiments.txt. This document is an adversarial-search study; the project has not introduced an economy.